The **Custom** tab lets you define **Metrics** using a mathematical function that you specify. These functions can include measures such as the sum of the number of hours of ozone exposure above 60 ppb. The possibilities are quite diverse, as evidenced by the range of functions and variables available for use as shown in Table 4-5. However, the syntax for using these functions is somewhat involved, so we have reserved discussion of this for the Appendix M: Function Editor.

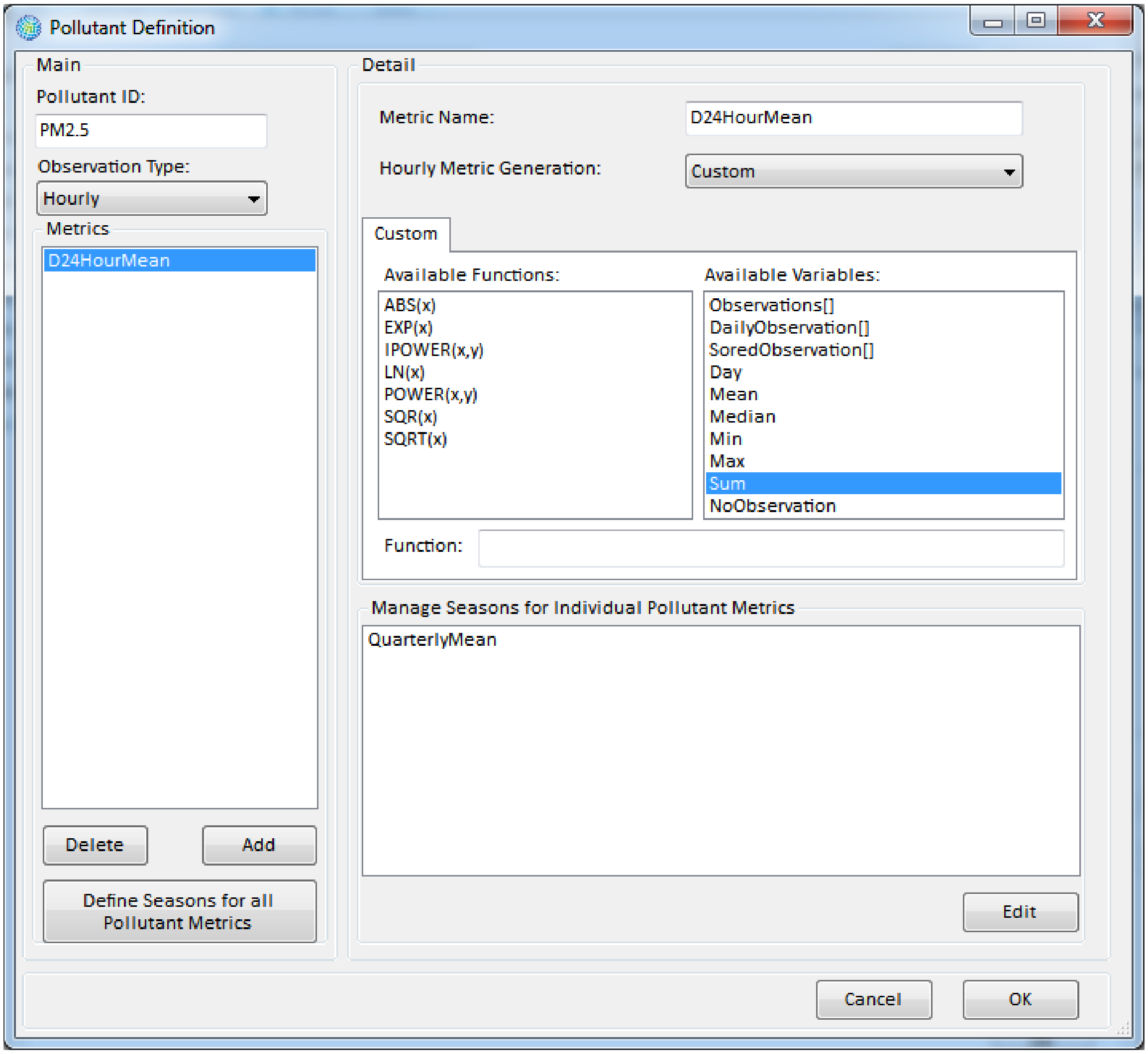


Table 4-5. Available Functions and Variables for Custom Metrics

Name	Description
Functions	
ABS(x)	Returns the absolute value of *x.*
EXP(x)	Returns *e* the power *x*, where *e* is the base of the natural logarithm.
IPOWER(x,y)	Returns *x* to the power *y* (*y* an integer value).
LN(x)	Returns the natural logarithm of *x.*
POWER(x,y)	Returns *x* to the power *y* (*y* a floating point value).
SQR(x)	Returns the square of *x*.
SQRT(x)	Returns the positive square root of *x*.
Variables	
Observations[i]	All hourly observations for the year (index begins at zero, typically ranging to 8,760).
DailyObservations[i]	All hourly observations for the day (indexed zero to twenty-three).
SortedObservations[i]	All hourly observations for the day, sorted from low to high (indexed zero to twenty-three).
Day	Index of the day whose metric value is being generated (index begins at zero).
Mean	Mean of the daily observations.
Median	Median of the daily observations
Min	Minimum of the daily observations.
Max	Maximum of the daily observations.
Sum	Sum of the daily observations.
NoObservation	Flag value indicating a missing observation (-345)

4.1.2.2 Manage Seasons for Individual Pollutant Metrics

Manage Seasons for Individual Pollutant Metrics allow you to aggregate daily **Metric** values over a portion of the year that you define. This has a number of uses. For example, if pollutant values vary greatly by season of the year, you can calculate separate pollutant measures for each season of interest. You might be interested in *Dry Season* versus *Wet Season*, or differences between *Winter*, *Spring*, *Summer*, and *Fall*.

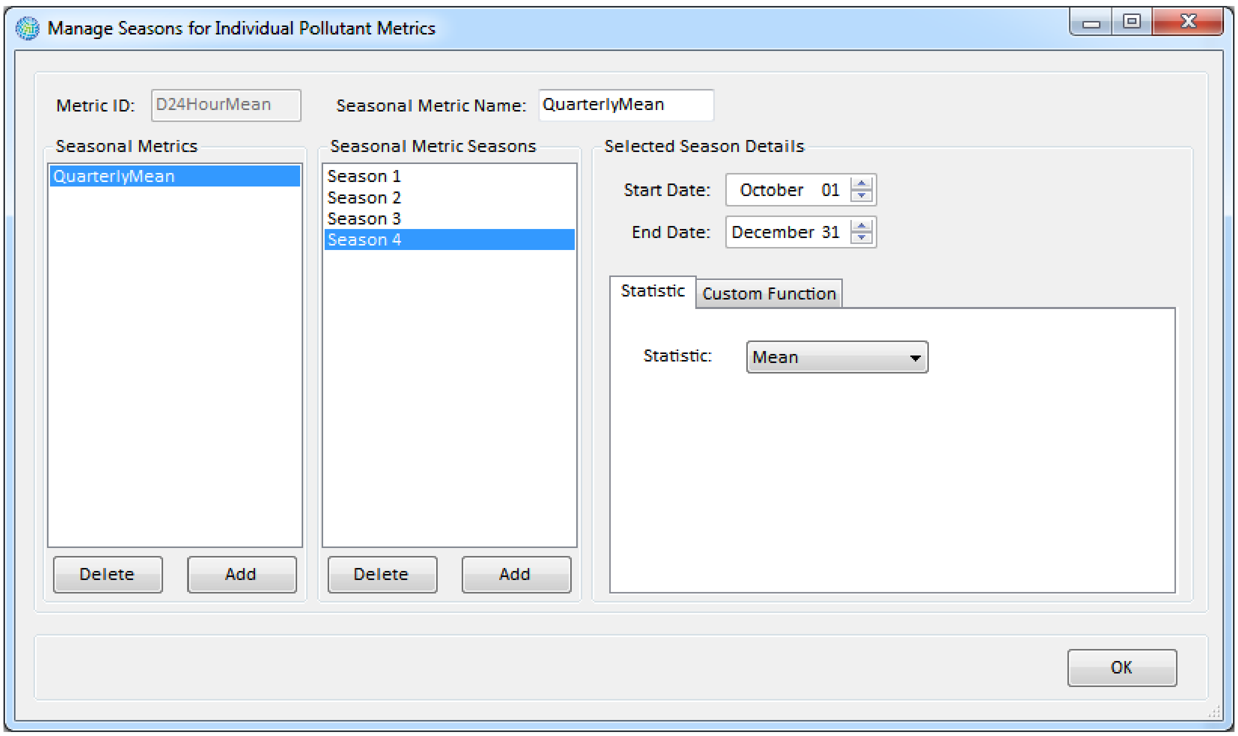
> **Fundamental Concept - Seasonal Metric**
>
> A **Seasonal Metric** allows you to define portions of the year over which to aggregate daily air quality metrics. For example, you could create a warm season and cold season for D8HourMax ozone and calculate an average WarmSeasonD8HourMax ozone value and an average ColdSeasonD8HourMax to use in an analysis of seasonal health impacts.

To add seasonal metrics for individual pollutant metrics, first select the metric of interest (e.g., D24HourMean), then click on the **Edit** button below the **Manage Seasons**

for Individual Pollutant Metrics box. The **Manage Seasons for Individual Pollutant Metrics** window will appear.

To add a **Seasonal Metric**, click on the **Add** button below the **Seasonal Metrics** box. A default name *Seasonal Metric 0* will appear in the box. Since the default name is not very descriptive, it is best to change the name to something more informative such as the *QuarterlyMean*. As with the **Metric** names, keep in mind that the **Seasonal Metric** names need to be consistent with the metric names that you include in your air pollution monitoring and modeling data, as well as your health impact functions.

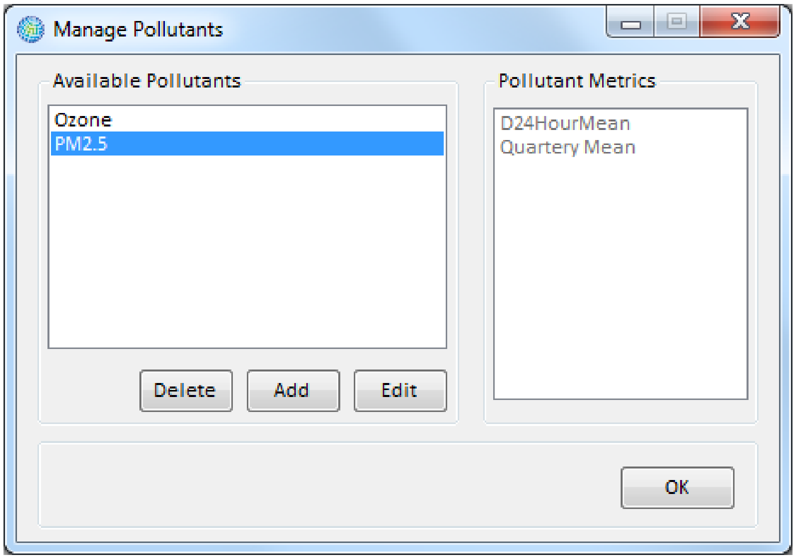
The next step is to define the seasons that you want associated with your **Seasonal Metric** name. For example, in the case of a *Quarterly Mean*, you would want to define four seasons. To start this process, click on the **Add** button below the **Seasonal Metric Seasons** box. Then, in the far right side of the window under **Selected Season Details**, give the **Start Date** and **End Date** for each season. To change the date, click on the month or day, and use the arrows to change the month or day accordingly.



Next, you need to choose the **Statistic** tab or the **Custom Function** tab to determine how the daily **Metrics** will be combined in each season. For example, you might choose the *Mean* from the drop-down list on the **Statistics** tab. This would calculate the mean of the daily metrics in each season. The **Custom** tab allows seasonal metric values to be calculated using customized functions, similar to those used to calculate daily metric values from hourly observations. See Appendix M: Function Editor for more detail on this topic.

Once you have finished defining the **Seasonal Metrics**, click **OK** to return to the **Pollutant Definition** window.

Click **OK** after defining each **Pollutant**. This will return you to the **Manage Pollutants** window.



4.1.2.3 Define Seasons for all Pollutant Metrics

The **Define Seasons for all Pollutant Metrics** button on the **Pollutant Definition** window allows you to associate **Seasons** with a **Pollutant** and all of its associated metrics. These seasons differ somewhat from the **Seasonal Metrics** defined for individual pollutant metrics (discussed above). They are used to define:

- The portion of the year for which benefits are calculated for a **Pollutant**. You can think of **Seasons** as defining this period of the year "globally" for the pollutant, as it affects the portion of the year over which both **Metrics** and **Seasonal Metrics** are calculated. For example, in the United States ozone benefits are often only calculated for the ozone season, from April 1 through September 30.

- The portion(s) of the year for which missing pollutant concentrations are filled in by BenMAP-CE. That is, in order to

> **Fundamental Concept – Global Season**
>
> Defining **Seasons** (or **Global Seasons**) for all pollutant metrics constrains a pollutant's benefits analyses to a user-specified portion of the year. For example, if you define the Global Season for ozone as April 1 – September 30, then BenMAP-CE will only calculate benefits during that window for each metric associated with ozone. Any Seasonal Metrics should be contained within and collectively span the Global Season. For example, for a Global Season of April 1 – Sept. 30, you might define a Seasonal Metric to analyze health impacts in early summer (April 1 – June 30) vs. late summer (July 1 – Sept 30).
>
> A Global Season also defines the portion of the year for which missing pollutant concentrations are estimated in by BenMAP-CE and how the replacement value is calculated (e.g., days with missing air quality data may be replaced with the average value from the Global Season).